

Robust Loss Functions for Treatment Effect Estimation

CATE Validation, R-Learning, and Policy Learning

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Machine Learning as Loss Minimization

- ▶ **General Definition:** A machine learning problem involves a **task**, a **loss function** (punishment for poor performance), and **data**.
- ▶ **Prediction (Standard Example):**
 - ▶ Task: Predict Y from X
 - ▶ Loss Function: Squared-error loss $L = (Y - \mu(X))^2$
 - ▶ Training: Minimize training loss under regularization
- ▶ **Inventory Management (Newsvendor Model):**
 - ▶ Task: Choose quantity Q to purchase at cost C and sell at price P
 - ▶ Loss Function: **Negative Profit**
 - ▶ Policy $\pi(X)$ is learned by minimizing the empirical estimate of negative value

Quiz 1: R-Loss Logic

What is the primary advantage of using the R-loss for estimating CATE?

1. It allows for simple T-learning without nuisance functions
2. It isolates the treatment effect by "partialling out" baseline variation and propensity via Robinson's transformation
3. It only works if the treatment effect is constant
4. It eliminates the need for cross-fitting

Quiz 1 Solution

1. Incorrect: Nuisance functions are still required
2. **Correct: It targets heterogeneity while remaining robust to confounding via residualization**
3. Incorrect: It is designed specifically to estimate the heterogeneous $\tau(x)$
4. Incorrect: Cross-fitting is still necessary for robustness

The R-Loss for CATE Estimation

- ▶ **Goal:** Frame CATE ($\tau(x)$) estimation as a loss minimization problem similar to prediction
- ▶ **Derivation:** From Robinson's transformation:

$$Y_i - m(X_i) = (W_i - e(X_i))\tau(X_i) + \varepsilon_i$$

- ▶ **The R-Loss (Targeting Loss):**

$$L(\tau(X_i); X_i, Y_i, W_i) = ((Y_i - m(X_i)) - \tau(X_i)(W_i - e(X_i)))^2$$

- ▶ This loss function allows us to use any ML algorithm (Lasso, Boosting, Neural Nets) to estimate $\tau(x)$ by simply swapping out the loss function

The R-Learner Strategy

- ▶ **The Problem:** The R-loss requires knowing the nuisance components $m(x)$ and $e(x)$
- ▶ **The Solution (R-Learner):**
 1. Fit $\hat{m}(x)$ and $\hat{e}(x)$ using supervised learning or regression approach
 2. Estimate $\tau(\cdot)$ by minimizing the cross-fit plug-in loss:

$$\hat{\tau}(\cdot) = \operatorname{argmin}_{\tau} \left\{ \frac{1}{n} \sum_{i=1}^n \left((Y_i - \hat{m}^{(-i)}(X_i)) - \tau(X_i)(W_i - \hat{e}^{(-i)}(X_i)) \right)^2 + \Lambda_n(\tau) \right\}$$

- ▶ **Quasi-Oracle Property:** For a large class of problems, the R-learner satisfies the same MSE bounds as if $m(x)$ and $e(x)$ were perfectly known

Forest-Based Methods and R-Loss Improvement

- **Empirical Results:** Forest-based methods can benefit from R-loss optimization

Method	T-forest	S-forest	X-forest	Causal Forest
Δ (R-loss improvement)	0.0040	0.0966	-0.0028	-0.0012
Std Err	0.0036	0.0030	0.0010	0.0006
t-stat	1.1084	32.3835	-2.8438	-2.0530

Quiz 2: Stacking Benefits

What is the main benefit of stacking multiple CATE estimators?

1. It reduces the need for cross-fitting
2. It combines the strengths of different models and can improve overall performance
3. It eliminates the need for propensity score estimation
4. It guarantees unbiased estimates

Quiz 2 Solution

1. Incorrect: Cross-fitting is still essential
2. **Correct: Stacking can leverage the strengths of different models**
3. Incorrect: Propensity scores are still needed for residualization
4. Incorrect: No method guarantees unbiased estimates in practice

Stacking Multiple CATE Estimators

- ▶ **Setup:** Suppose we have multiple candidate CATE estimators $\hat{\tau}_1, \dots, \hat{\tau}_M$
- ▶ **Combining Estimators:** Form a convex combination:

$$\hat{\tau}_\alpha(x) = \sum_{m=1}^M \alpha_m \hat{\tau}_m(x), \quad \alpha_m \geq 0, \quad \sum_m \alpha_m = 1$$

- ▶ **Optimization:** Choose α by minimizing the R-loss on held-out data:

$$\min_{\alpha \geq 0, \mathbf{1}^\top \alpha = 1} \frac{1}{n} \sum_i \left(\tilde{Y}_i - \hat{\tau}_\alpha(X_i) \tilde{W}_i \right)^2$$

- ▶ **Result:** This is a quadratic program that can be solved with standard convex optimization tools

Calibration Checks and QINI Curves

- ▶ **Calibration:** Regress outcomes on predicted CATE to see if the model under- or over-fits. We want a slope $\beta \approx 1$
- ▶ **QINI Curve:** Evaluates the cost-benefit of prioritizing units by their CATE ranking
 - ▶ **X-axis:** Fraction of units treated (cost)
 - ▶ **Y-axis:** Cumulative gain from treating the top X fraction

How is the Qini Score Calculated?

Formula

$$\text{Qini}(\pi) = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)) \cdot \pi(X_i) - \text{Baseline Gain}$$

Definitions:

- ▶ $Y_i(1)$ = Outcome if treated
- ▶ $Y_i(0)$ = Outcome if untreated
- ▶ $\pi(X_i)$ = Policy decision (1 = treat, 0 = control)
- ▶ **Baseline Gain** = Expected gain from treating everyone or no one (usually 0 if comparing to treating no one)

Intuition:

- ▶ Measures **extra value** generated by the policy compared to random treatment
- ▶ Higher Qini = more effective targeting of high-impact individuals
- ▶ Used to find the optimal treatment fraction

Policy Learning vs. CATE Estimation

Quiz 3: Policy Value

If the optimal policy $\pi(x)$ thresholds CATE at a value C , what does C represent?

1. The average treatment effect
2. The propensity score
3. The cost of the treatment
4. The standard error of the estimate

Quiz 3 Solution

1. Incorrect
2. Incorrect
3. **Correct: In general, we treat if $\tau(x) > C$, where C is the cost/threshold for treatment**
4. Incorrect

Quiz 4: Policy Learning Objective

What is the primary objective in policy learning?

1. To estimate the average treatment effect
2. To maximize the utilitarian value $V(\pi) = \mathbb{E}[Y_i(\pi(X_i))]$
3. To minimize the variance of the CATE estimates
4. To find the propensity score

Quiz 4 Solution

1. Incorrect
2. **Correct: The goal is to find a decision rule that maximizes expected outcome**
3. Incorrect
4. Incorrect

Policy Learning vs. CATE Estimation

- ▶ **CATE Estimation (Descriptive):** How much does the treatment affect unit i ?
- ▶ **Policy Learning (Prescriptive):** *Should* we treat unit i ? ($\pi : X \rightarrow \{0, 1\}$)
- ▶ **Goal:** Maximize the utilitarian/aggregate value $V(\pi) = \mathbb{E}[Y_i(\pi(X_i))]$
- ▶ **The Value Loss:** A good loss function should capture negative value (regret)

Descriptive vs. Prescriptive Causal Analysis

Descriptive (CATE Estimation)

- ▶ **Goal:** Estimate *how much* treatment affects each unit
- ▶ **Focus:** Individual treatment effects:
 $\tau_i = Y_i(1) - Y_i(0)$
- ▶ **Method:** Predict Conditional Average Treatment Effects (CATE)
- ▶ **Output:** $\hat{\tau}(x_i)$ for each individual
- ▶ **Question Answered:** "What is the effect of treatment on this person?"
- ▶ **Example:** Estimating how much a drug affects each patient's blood pressure

Prescriptive (Policy Learning)

- ▶ **Goal:** Determine *whether* to treat each unit
- ▶ **Focus:** Optimal treatment policy:
 $\pi : X \rightarrow \{0, 1\}$
- ▶ **Method:** Maximize expected outcome $V(\pi) = \mathbb{E}[Y(\pi(X))]$
- ▶ **Output:** Binary decision rule $\pi(x_i)$ for each individual
- ▶ **Question Answered:** "Should we treat this person?"
- ▶ **Example:** Deciding which patients should receive a medical treatment

IPW vs. AIPW in Policy Learning

IPW (Inverse Propensity Weighting)

- ▶ **Meaning:** Causal technique that reweights observations by inverse probability of actual treatment
- ▶ **Purpose:** Corrects selection bias by upweighting underrepresented treatment groups
- ▶ **In Policy Learning:** Estimates policy value by weighting outcomes inversely to propensity scores

Key Difference

- ▶ **IPW:** Relies solely on propensity scores
- ▶ **AIPW:** Augments IPW with outcome modeling, making it robust to model misspecification

AIPW (Augmented IPW)

- ▶ **Meaning:** Improved IPW that combines propensity weighting with outcome regression (doubly robust)
- ▶ **Purpose:** Reduces variance and increases robustness using both treatment and outcome models
- ▶ **In Policy Learning:** Enhances IPW loss with outcome predictions for stable, efficient evaluation

IPW and AIPW Policy Losses

- ▶ **IPW Loss:** Learns a policy by maximizing an inverse-propensity weighted estimate of value
- ▶ **AIPW Loss:** Improves on IPW by adding outcome augmentation
- ▶ **The Objective:** Maximize $\hat{V}(\pi) = \frac{1}{n} \sum (2\pi(X_i) - 1)\hat{\Gamma}_i$
- ▶ **Result:** This reduces to a **weighted classification problem**, where we classify individuals into "treat" or "control" based on their double-robust scores $\hat{\Gamma}_i$

Case Study: California GAIN Study

- ▶ Job training program evaluation across four counties
- ▶ By optimizing over a class of low-depth trees, we learn a policy that provides significantly more utility than random assignment
- ▶ **Ethics Note:** Policies can be constrained to *not* split on protected variables (e.g., race, gender) even if they exhibit heterogeneity

Key Takeaways

- ▶ **The R-Loss** allows CATE estimation to be treated as a standard ML problem using Robinson's transformation
- ▶ **R-Learners** use cross-fitted nuisance components to achieve quasi-oracle performance
- ▶ **Validation** of CATE is done via R-loss differences, calibration tests, and QINI curves
- ▶ **Policy Learning** focuses on prescriptive utility maximization, often using Doubly Robust scores to solve a weighted classification problem
- ▶ **Stacking experts** provides a powerful way to combine multiple CATE estimators

Summary Table: Loss Functions Comparison

Method	Strengths	Limitations
R-Loss	Doubly robust, targets CATE directly, works with any ML algorithm	Requires nuisance estimation, needs cross-fitting
IPW Policy	Simple, works with any classifier	High variance, not doubly robust
AIPW Policy	Doubly robust, semi-parametrically efficient	More complex, needs both nuisance estimates
Stacking	Combines strengths of multiple models	Computationally intensive, needs diverse models